

Problem 1 : Oscillation Function

Given a bounded function $f : \mathbb{R} \rightarrow \mathbb{R}$, define the oscillation function of f by

$$\omega_f(x) = \lim_{\delta \rightarrow 0^+} \sup\{|f(y) - f(z)| : y, z \in (x - \delta, x + \delta)\}, \quad x \in \mathbb{R}.$$

(a) Show that the limit in the definition above exists and that f is continuous at the point x precisely when $\omega_f(x) = 0$.

(b) Show that the function $x \mapsto \omega_f(x)$ is upper semi-continuous in the sense that the set

$$\{x \in \mathbb{R} : \omega_f(x) < A\}$$

is open, for each $A \in \mathbb{R}$.

(c) **(Basic Exam, Spring 2025)** Prove that the function f cannot be continuous on the set $\mathbb{Q}$ of rational numbers only.

(d) Does there exist a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous on the set of irrational numbers only? Prove your assertion.

Solution. (a): Let $g_x(\delta) := \sup\{|f(y) - f(z)| : y, z \in (x - \delta, x + \delta)\}$. Note that for $\delta_1 < \delta_2$, $(x - \delta_1, x + \delta_1) \subset (x - \delta_2, x + \delta_2) \Rightarrow \{|f(y) - f(z)| : y, z \in (x - \delta_1, x + \delta_1)\} \subset \{|f(y) - f(z)| : y, z \in (x - \delta_2, x + \delta_2)\}$ thus $g(\delta_1) \leq g(\delta_2)$. So g is nonincreasing as $\delta \rightarrow 0$. Furthermore $g_x \geq 0$. Thus the limit exists.

Let $C(f) := \{x \in \mathbb{R} : f \text{ continuous at } x\}$. If $x \in C(f)$, $\forall \varepsilon > 0$ there exists $\delta > 0$ such that for all $z, y \in (x - \delta, x + \delta)$ $|f(x) - f(y)| < \frac{\varepsilon}{2}$. Thus

$$\begin{aligned} |f(y) - f(z)| &\leq |f(y) - f(x)| + |f(x) - f(z)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus $g_x(\delta) < \varepsilon$. Thus $\omega_f(x) = 0$. Suppose $\omega_f(x) = 0$. Then $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\delta_0 \in [0, \delta]$ $g_x(\delta_0) < \varepsilon$. Thus for all $y, z \in (x - \delta, x + \delta)$, $|f(y) - f(z)| < \varepsilon$ it immediately follows that $|f(x) - f(y)| < \varepsilon$. Thus $x \in C(f)$.

(b): $\forall A \in \mathbb{R}$, let $U_A := \{x \in \mathbb{R} : \omega_f(x) < A\}$. $\forall x \in U_A$, we know $\forall \varepsilon > 0$ there exists a $\delta > 0$ where for $y, z \in (x - \delta, x + \delta)$ we have $|f(y) - f(z)| < A$. For $y \in (x - \delta, x + \delta)$, take $\delta_0 < \delta - |y - x|$ for $z \in (y - \delta_0, y + \delta_0)$

$$|z - x| \leq |z - y| + |y - x| < \delta_0 + |y - x| < \delta.$$

Thus

$$|f(z_1) - f(z_2)| < A$$

for $z_1, z_2 \in (y - \delta_0, y + \delta_0) \subset (x - \delta, x + \delta)$. Thus $(x - \delta, x + \delta) \subset U_A$. Thus U_A is open and ω_f is upper semicontinuous.

(c):

$$C(f) = \{x \in \mathbb{R} : \omega_f(x) = 0\} = \bigcap_{n \in \mathbb{N}} U_{\frac{1}{n}}.$$

Assume the contrary that $\mathbb{Q} = U_0$, thus $\mathbb{Q} \in U_{\frac{1}{n}}$ thus the sets are dense. By above they are also open. Note for all $x \in U_{\frac{1}{n}}$, $U_{\frac{1}{n}} - \{x\}$ is still dense and open. Enumerate $\mathbb{Q}$ and let q_i be the corresponding sequence. By Baire Category Theorem, $\emptyset = \bigcap_{n \in \mathbb{N}} (U_{\frac{1}{n}} - \{q_n\})$ is a dense set which is a contradiction. Thus $C(f) \neq \mathbb{Q}$

Problem 2 : Uniform Continuity and Extension

(a) Let X and Y be metric spaces, with Y complete, and let $A \subseteq X$ be dense in X . Let $f : A \rightarrow Y$ be a uniformly continuous map. Show that f can be extended uniquely to a continuous map $\tilde{f} : X \rightarrow Y$. Show also that the unique extension $\tilde{f}$ is uniformly continuous on X .

(b) (**Basic Exam, Spring 2024**) Let $f : \mathbb{Q} \rightarrow \mathbb{R}$ be a uniformly continuous function. Show that the closure $\overline{f(\mathbb{Q})}$ is connected.

Solution. (a): $\forall x \in X - A$. Create a sequence where $a_n \in A \cap B_{\frac{1}{n}}(x)$ where the intersection is nonempty because of density of A in X . Note $a_n \rightarrow x$ and is hence Cauchy. By the sequential definition of uniform continuity, a_n being Cauchy $\Rightarrow f(a_n)$ is Cauchy. By completeness of Y , $\lim f(a_n)$ exists. We want to define $f(x) = \lim f(a_n)$ but we need to show that this definition doesn't rely on the choice of sequence.

Let $a_n \rightarrow x$ and $b_n \rightarrow x$ be sequences in A . Define c_n , where $c_{2n-1} = a_n$ and $c_{2n} = b_n$, thus $c_n \rightarrow x$ as well. By uniform convergence and Completeness, $f(a_n) \rightarrow L \in Y$ and $f(b_n) \rightarrow L' \in Y$ and $f(c_n) \rightarrow \lim f(c_n) \in Y$. Definition of convergence and the fact that $f(a_n)$ and $f(b_n)$ are subsequences of $f(c_n)$, we know $L = \lim_{n \rightarrow \infty} f(c_n) = L'$. Hence the limit is well defined and is not dependent on the choice of sequence in A .

Thus let $F : X \rightarrow Y$ where

$$F(x) := \lim_{n \rightarrow \infty} f(a_n).$$

$\forall \varepsilon > 0: \exists \delta > 0$ such that $\forall a \in B_\delta(x) \cap A$ where $|F(x) - F(a)| = |F(x) - f(a)| < \frac{\varepsilon}{2}$.

$|F(x) - F(y)| \leq |F(x) - F(a)| + |F(y) - F(a)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Problem 3 : Compactness via Approximation

Let A be a closed subset of a complete metric space (X, d) . Assume that for each $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subseteq X$ such that for all $x \in A$, we have

$$\inf_{y \in K_\varepsilon} d(x, y) < \varepsilon.$$

Show that A is compact.

Solution. We will show this by showing A is complete and totally bounded. A is a closed subset of a complete metric space and hence complete (because it contains all its limit points). Now the hard part is showing A is totally bounded.

For all $\varepsilon > 0$, we have $K_{\frac{\varepsilon}{2}}$ a compact set where $\inf_{y \in K_{\frac{\varepsilon}{2}}} d(a, y)$ is achieved and is less than $\frac{\varepsilon}{2}$. Thus $d(y, K_{\frac{\varepsilon}{2}}) \leq \frac{\varepsilon}{2}$ for all $y \in A$. Thus $\exists b_1, \dots, b_n \in K_{\frac{\varepsilon}{2}}$ such that $\cup_{i=1}^n B_{\frac{\varepsilon}{2}}(b_i) = K_{\frac{\varepsilon}{2}}$. $\forall a \in A$, $\exists k \in K_{\frac{\varepsilon}{2}}$ such that $d(a, k) < \frac{\varepsilon}{2}$. Furthermore, since $K_{\frac{\varepsilon}{2}}$ is covered by $\frac{\varepsilon}{2}$ balls, there exists $b_i \in \{b_1, \dots, b_n\}$ such that $d(k, b_i) = \frac{\varepsilon}{2}$. Thus

$$d(b_i, a) \leq d(b_i, k) + d(k, a) \leq \varepsilon.$$

Thus

$$A \subset \bigcup_{i=1}^n B_{\varepsilon}(b_i).$$

We now need to show that we can perturb the balls sufficiently and move them into A .

Pick a $a_i \in A \cap B_{\varepsilon}(b_i)$, $\forall a \in A \cap B_{\varepsilon}(b_i)$:

$$d(a_i, a) \leq d(a_i, b_i) + d(b_i, a) < 2\varepsilon.$$

Thus

$$A \subset \bigcup_{i=1}^n B_{2\varepsilon}(a_i)$$

Thus A totally bounded. A is compact.

Problem 4 : l^∞ as a Complete Metric Space

Let $l^\infty = l^\infty(\mathbb{N})$ denote the set of all bounded complex-valued sequences $x = (x_n)_{n=1}^\infty$.

(a) Show that l^∞ is a complete metric space when equipped with the metric

$$d(x, y) = \sup_n |x_n - y_n|.$$

(b) Let $(\varepsilon_n)_{n=1}^\infty$ be a sequence of positive real numbers such that $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. Show that

$$M = \{x = (x_n) \in l^\infty : |x_n| \leq \varepsilon_n, \quad n = 1, 2, \dots\}$$

is a compact subset of l^∞ .

Solution.